Modeling of Hydraulic (Fluid) Systems – Part 1

(流体系统的建模1)

Main Contents of Course (课程主要内容)

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Electrical (电路)

Electro-Mechanical (机–电)

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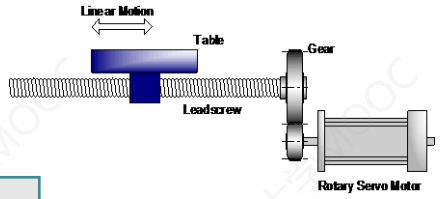
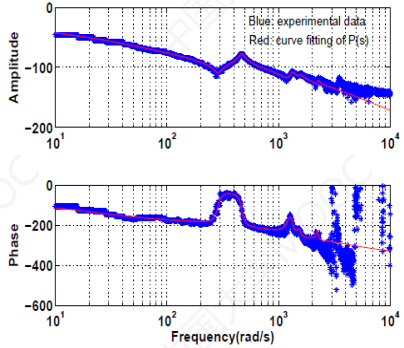
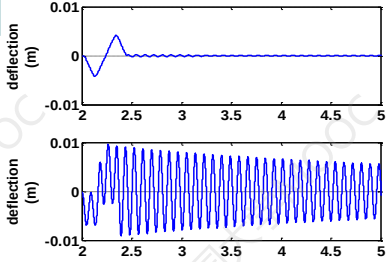






Variables

- q : volumetric flow rate (体积流量) **[m³/sec]** (current)
- V: volume (体积) **[m³]** (charge)
- p : pressure (压力) **[N/m²]** (voltage)

The analogy between the hydraulic system and the electrical system will be used often (经常将液压系统和电路系统作相似).

•Just as in electrical systems, the flow rate (current) is defined to be the time rate of change (derivative) of volume (charge) (与电路系统类似，流量被定义成体积的时间变化率):

$$q = \frac{d}{dt}V = \dot{V}$$

•The pressure, p, is the *absolute pressure* (压力是指绝对压力).

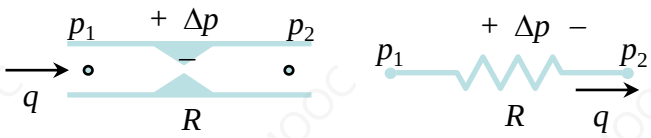
Be careful in determining whether the pressure is the absolute pressure or *gauge pressure*, p^* (注意所定义的压力是绝对压力还是表压). Gauge pressure is the difference between the absolute pressure and the atmospheric pressure, i.e (表压是指绝对压力和大气压力的差值).

$$p^* = p - p_{atmospheric}$$

Basic Modeling Elements

- **Fluid Resistance (液阻)**

  Describes any physical element with the characteristic that the pressure drop, Δp , across the element is proportional to the flow rate, q. (表示物理元件两端压降对流量的比值)



$$\Delta p = p_1 - p_2 = p_{12} = R \cdot q$$

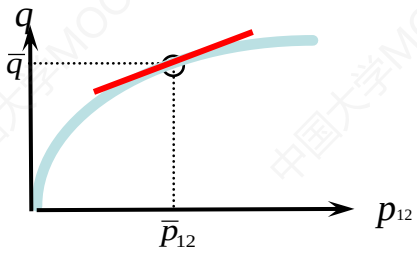
$$q = \frac{1}{R}\Delta p = \frac{1}{R} p_{12}$$

– Orifices, valves, nozzles and friction in pipes can be modeled as fluid resistors. (节流口、阀、喷嘴和管道中的摩擦等都能被建模成液阻)

Ex: *The flow that goes through an orifice or a valve and the turbulent flow that goes through a pipe is related to the pressure drop by* (经过节流口或阀的流量和经过管道的紊流流量与压降有关)

$$q = k\sqrt{p_{12}}$$

Find the effective flow resistance of the element at certain operating point ($\bar{q}$, $\bar{p}_{12}$) (找出元件在某工作点下的有效液阻).



$$\tilde{q} = \frac{dq}{dp_{12}}\tilde{p}_{12}$$

$$\tilde{q} = q - \bar{q},$$

$$\tilde{p}_{12} = p_{12} - \bar{p}_{12}$$

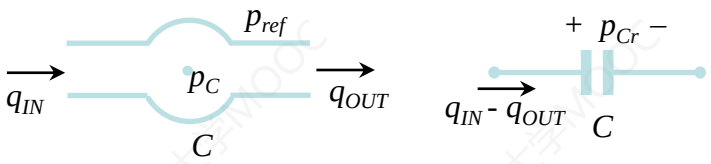
$$\frac{1}{R} = \left.\frac{dq}{dp_{12}}\right|_{(\bar{q},\bar{p}_{12})} = \frac{k}{2\sqrt{\bar{p}_{12}}}$$

$$R = \frac{2\sqrt{\bar{p}_{12}}}{k} = \frac{2\bar{q}}{k^2}$$

Basic Modeling Elements

- **Fluid Capacitance (液容)**

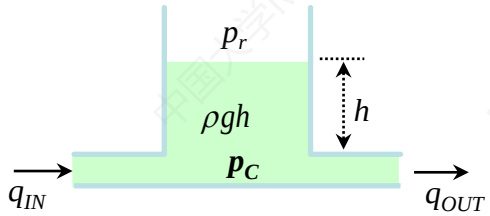
  Describes any physical element with the characteristic that the rate of change in pressure, p, in the element is proportional to the difference between the input flow rate, q_{IN} , and the output flow rate, q_{OUT} .(表示物理元件两端压力变化率对输入输出流量之差的比值)



$$C\frac{d}{dt}\underbrace{\left(p_C - p_{ref}\right)}_{p_{Cr}} = C\cdot\dot{p}_{Cr} = q_{IN} - q_{OUT}$$

– Hydraulic cylinder chambers, tanks, and accumulators are examples of fluid capacitors (液压缸、油箱和蓄能器可看作是液容)

Ex: *Consider an open tank with a constant cross-sectional area, A* (考虑具有定常横截面积*A*的敞开式油箱):



$$p_C = p_r + \rho gh \Rightarrow p_{cr} = \rho gh$$

$$q_{IN} - q_{OUT} = \frac{d}{dt}(V_{volume}) = \frac{d}{dt}(Ah) = A\frac{d}{dt}h = A\dot{h}$$

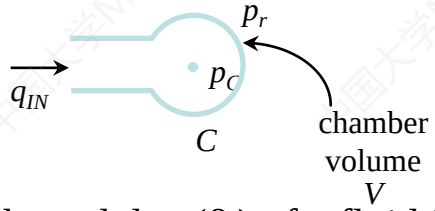
$$\dot{p}_{Cr} = \rho g\dot{h} = \frac{\rho g}{A}(q_{IN} - q_{OUT})$$

$$\Rightarrow$$

$$C = \frac{q_{IN} - q_{OUT}}{\dot{p}_{Cr}} = \frac{A\dot{h}}{\rho g\dot{h}} = \frac{A}{\rho g}$$

Fluid Capacitance Examples

Ex: Calculate the equivalent fluid capacitance for a hydraulic chamber with only an inlet port **(计算具有1个进口的液压容腔的等效液容).**



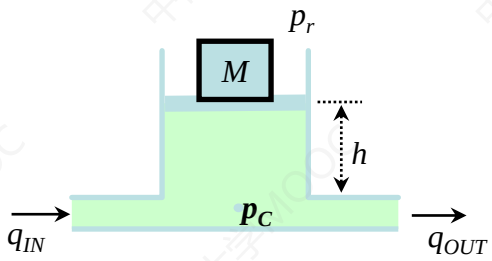
Recall the bulk modulus (β) of a fluid is defined by **(回顾流体的弹性模量定义):**

$$\beta = V\frac{dp_{Cr}}{dV} = V\frac{\left(dp_{Cr}/dt\right)}{\left(dV/dt\right)} = V\frac{\dot{p}_{Cr}}{q_{in}}$$

$$\Rightarrow$$

$$q_{IN} = \underbrace{\left(\frac{V}{\beta}\right)}_{C}\frac{d}{dt}p_{Cr}$$

Ex: Will the effective capacitance change if in the previous open tank example, a load mass M is floating on top of the tank ? **(如下所示的敞开式油箱顶部有一浮动的负载质量*M*，其有效液容会改变吗)**

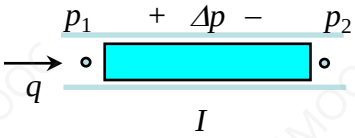


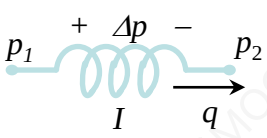
$$\Rightarrow \quad C = \frac{q_{IN} - q_{OUT}}{\dot{p}_{Cr}} = \frac{A}{\rho g}$$

Basic Modeling Elements

- **Fluid Inertance (Inductance) (液感)**

Describes any physical element with the characteristic that the pressure drop, Δp , across the element is proportional to the rate of change (derivative) of the flow rate, q. (表示物理元件两端的压降对流量变化率的比值)





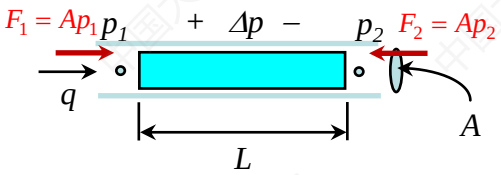
$$\Delta p = p_{12} = (p_1 - p_2) = I\frac{d}{dt}q = I \cdot \dot{q}$$

– Long pipes are examples of fluid inertance (长管道可看作是流体惯性阻力，即液感).

Q: What will happen if you suddenly shut off one end of a long tube? (长管道一端突然关闭会发生什么？)

--- (Water Hammer effect) (水锤效应)

Ex: Consider a section of pipe with cross-sectional area A and length L, filled with fluid whose density is ρ (考虑具有定常截面积A和长度L，液体密度为ρ的一端管道):



Start with force balance (力平衡): $F = ma$

$$\begin{cases} F = F_1 - F_2 = Ap_1 - Ap_2 = Ap_{12} \\ m = \rho AL \\ a = \frac{d}{dt}(v) = \frac{d}{dt}(\frac{q}{A}) = \frac{\dot{q}}{A} \end{cases}$$

$\Rightarrow$

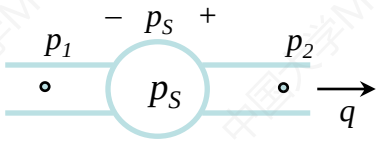
$$Ap_{12} = \rho AL.\frac{1}{A}\dot{q} \quad \Rightarrow \quad I = \frac{\rho L}{A}$$

$$p_{12} = \underbrace{\frac{\rho L}{A}}_{I}\dot{q}$$

Basic Modeling Elements

- **Pressure Source (Pump) (恒压泵)**
  - An ideal pressure source of a hydraulic system is capable of maintaining the desired pressure, regardless of the flow required for what it is driving **(液压系统的理想压力源能维持期望的压力，而不管所需的驱动流量).**

Voltage Source **(电压源)**



$$p_{21} = p_2 - p_1 = p_S$$

- **Flow Source (Pump)（恒流泵）**
  - An ideal flow source is capable of delivering the desired flow rate, regardless of the pressure required to drive the load **(液压系统的理想流量源能传递期望的流量，而不管所需的驱动压力).**

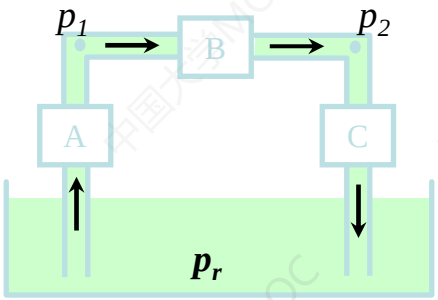
Current Source **(电流源)**

$$q = q_S$$

Interconnection Laws

- **Compatibility Law (相容性定律)**
  - The sum of the pressure drops around a loop must be zero **(回路中的压降总和为零)**.
  - Similar to the Kirchhoff's voltage law **(类似于基尔霍夫电压定律)**.

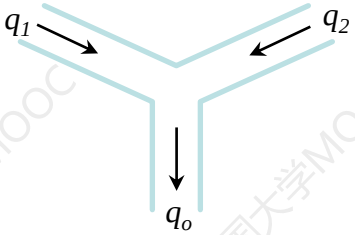
$$\sum_{\substack{\text{Closed}\\\text{Loop}}} \Delta p_j = \sum_{\substack{\text{Closed}\\\text{Loop}}} p_{ij} = 0$$



- **Continuity Law (连续性定律)**
  - The algebraic sum of the flow rates at any junction in the loop is zero **(回路中任意连接点的流量代数和为零)**.
  - This is the consequence of the conservation of mass **(质量守恒性的推论)**.
  - Similar to the Kirchhoff's current law law **(类似于基尔霍夫电流定律)**.

$$\sum_{\substack{\text{Any}\\\text{Node}}} q_j = 0$$

$$\text{or} \quad \sum q_{IN} = \sum q_{OUT}$$



Summary

- **Basic Modeling Elements (基本建模元件)**
  - Resistance (液阻)
  - Capacitance (液容)
  - Inertance (液感)
  - Pressure and Flow Sources (恒压泵和恒流泵)
- **Interconnection Relationships (互联关系)**
  - Compatibility Law (相容性定律)
  - Continuity Law (连续性定律)